# Appendix E - Music Note and Frequency Tables

Each audio engine programs pitch through either a direct Hertz value, a period/divisor, or a chip-specific frequency register. This appendix gives the formula for each engine and a 12-note octave-4 table.

The reference frequencies are:

- **SoundChip / SFX / WAV / MOD / Amiga Paula DMA:** programmed in Hertz directly (the mixer runs at the system output sample rate). No divisor formula applies.
- **PSG / AY-3-8910:** master clock `2,000,000` Hz, period `clock / (16 * f)`.
- **SN76489:** master clock `3,579,545` Hz (NTSC), period `clock / (32 * f)`.
- **SID:** master clock `985,248` Hz (PAL), period `f * 16777216 / clock`.
- **TED audio:** PAL master clock `886,724` Hz, sound clock `110,840` Hz, register `1024 - sound_clock / f`.
- **POKEY:** master clock `1,789,773` Hz (NTSC) divided by `28`, divisor `clock / 28 / f - 1`.
- **MIDI/MUS:** use MIDI note numbers. The player converts note `69` to A4 = `440` Hz and applies pitch bend internally.

## E.1 Octave 4 (middle C through B), equal temperament A4 = 440 Hz

| Note | f (Hz) | PSG period | SN76489 period | SID period | TED register | POKEY divisor |
|---|---|---|---|---|---|---|
| C | 261.63 | 478 | 427 | 4455 | 600 | 243 |
| C# | 277.18 | 451 | 403 | 4720 | 624 | 229 |
| D | 293.66 | 426 | 381 | 5001 | 647 | 216 |
| D# | 311.13 | 402 | 359 | 5298 | 668 | 204 |
| E | 329.63 | 379 | 339 | 5612 | 688 | 192 |
| F | 349.23 | 358 | 320 | 5947 | 707 | 182 |
| F# | 369.99 | 338 | 302 | 6300 | 724 | 171 |
| G | 392.00 | 319 | 285 | 6675 | 741 | 162 |
| G# | 415.30 | 301 | 269 | 7073 | 757 | 153 |
| A | 440.00 | 284 | 254 | 7493 | 772 | 144 |
| A# | 466.16 | 268 | 240 | 7939 | 786 | 136 |
| B | 493.88 | 253 | 226 | 8410 | 800 | 128 |

## E.2 Extending the table

To move up one octave (double the frequency): halve the PSG and SN76489 periods, double the SID period, halve the POKEY divisor, and for TED halve the internal divisor:

```
newTED = 1024 - INT((1024 - oldTED) / 2)
```

To move down one octave (halve the frequency): double the PSG and SN76489 periods, halve the SID period, double the POKEY divisor, and for TED double the internal divisor:

```
newTED = 1024 - 2 * (1024 - oldTED)
```

The relation is exact to within rounding error. Octave shifts of more than four become noticeably flat or sharp on the small-divisor chips (SN76489, POKEY, and high TED values) and need a tempered correction table for accurate music.

## E.3 SoundChip, MIDI/MUS, and modern engines

The SoundChip channels and the sample-based engines (WAV, MOD, SFX, Amiga Paula DMA) take frequency in Hertz directly through the `FREQ` register or period model owned by that engine. A program writes `int(round(f))` into a SoundChip `FREQ` register and the channel plays at that frequency. No table is needed: middle C is `262`, A4 is `440`.

MIDI/MUS does not expose note pitch as a divider register. The file stores note numbers, programme changes, volume/expression controllers, tempo, and pitch bend. The RawlandMini synth converts those note numbers to frequency internally, with note `60` as middle C and note `69` as A4.

## E.4 Tuning notes

The numbers in section E.1 are calculated, not measured. The real-silicon SID, PSG, and POKEY chips drift with temperature and with the precise master-clock crystal; an Intuition Engine program that needs the same pitch on every machine should derive its own period from the engine clock at startup rather than hard- coding the values above.

The reference table assumes equal temperament. A program that wants just intonation, meantone, or other historical tunings generates the period table from the desired ratios using the same divisor formula. The engines have no awareness of a "scale": each channel plays whatever pitch its period register encodes.